

Bhargav Singapuri

AI Researcher | Seeking Summer 2026 Internship | Available May – August 2026
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PROFESSIONAL SUMMARY

AI researcher focused on mechanistic interpretability and LLM safety. Currently investigating transliteration as a jailbreak attack vector on Gemma 3 and building mitigations using linear probes and sparse autoencoders (SAEs). Benchmarking cross-coders for binary decision tasks. Hands-on with TransformerLens, SAEs, PyTorch, and Hugging Face. MS in AI at Northeastern; BS in Data Science & AI from NTU Singapore.

EDUCATION

Northeastern University

Master of Science in Artificial Intelligence | GPA: 3.83

Boston, MA

September 2024 – December 2026 (Expected)

Nanyang Technological University

Bachelor of Science in Data Science and Artificial Intelligence | GPA: 3.73

Singapore

August 2019 – July 2023

PROJECTS

Hidden in Plain Sight

January 2026 – Present

- Investigating how transliteration of harmful prompts (e.g. rendering English in Devanagari script) bypasses safety filters in Gemma 3
- Building detection mitigations using linear probes and sparse autoencoders

Cross-coder Interpretability

January 2026 – Present

- Evaluating cross-coders as a practitioner-facing interpretability tool on binary classification tasks, benchmarking against SAE baselines

Cancer Cell Detection

January 2025 – May 2025

- Fine-tuned Vision Transformers on lung-cell imagery using PyTorch Lightning; achieved 99% accuracy / 0.99 F1
- Quantified uncertainty with Monte-Carlo Dropout to ensure reliability in clinical inference scenarios

JournalGPT

September 2024 - October 2024

- Created a Retrieval-Augmented Generation (RAG) pipeline via Flowise, enabling semantic querying of personal journal entries with conversational tone-matching

SKILLS

Specialisations

Natural Language Processing (LLMs, RAG, Prompt Engineering, Agents), Computer Vision (CNNs, ViTs), Model Interpretability, Monte-Carlo Simulation, Data Visualization

ML/AI Frameworks

PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, LangChain, OpenCV, TransformerLens, SAEs

Data & Analytics

SQL (PostgreSQL, Snowflake), Pandas, NumPy, Airflow, Tableau, Excel (PowerQuery & Pivot Tables)

MLOps & Deployment

Docker, LangSmith, Terraform, Airflow, REST APIs, CI/CD Pipelines

Cloud & Infrastructure

AWS, Google Cloud Platform (GCP), Azure

Programming & Tools

Python, Shell Scripting, Git, Linux/UNIX, CI/CD

EXPERIENCE

Biogen

Boston, MA

Machine Learning Co-op

July 2025 – December 2025

- Engineering Enablement:** Authored and standardized an internal AI-assisted development playbook for Claude Code and OpenAI Codex, adopted by 10+ engineers, reducing prototyping time and accelerating delivery of features across 3 workflows
- Knowledge Deduplication:** Implemented ML pipelines to detect redundant SOPs via semantic clustering, reducing internal documentation load by 10%

United Overseas Bank

Singapore

Data Engineer

July 2023 – August 2024

- Dashboarding:** Spearheaded a Splunk dashboard visualizing TPS, latency, and HTTP statuses on 10M+ daily API records, cutting incident diagnosis time from 10 hours to 10 seconds
- Infrastructure Reliability:** Collaborated with cross-functional teams to deliver database upgrades, ensuring minimal downtime and security of customer data

Vertex Holdings

Singapore

Data Science Intern

December 2021 – April 2023

- Data Pipeline Engineering:** Developed Airflow ETL pipelines ingesting 1M+ relationship and investment records into Snowflake, creating a unified data layer that improved accessibility of investor intelligence for investment analysis and deal sourcing
- Investor Relationship Platform:** Built a CRM that centralized data on investment and business-development opportunities, generating relationship graphs and a searchable knowledge base for investment managers to map investor networks and automatically log outreach, eliminating manual reconciliation and speeding up deal flow